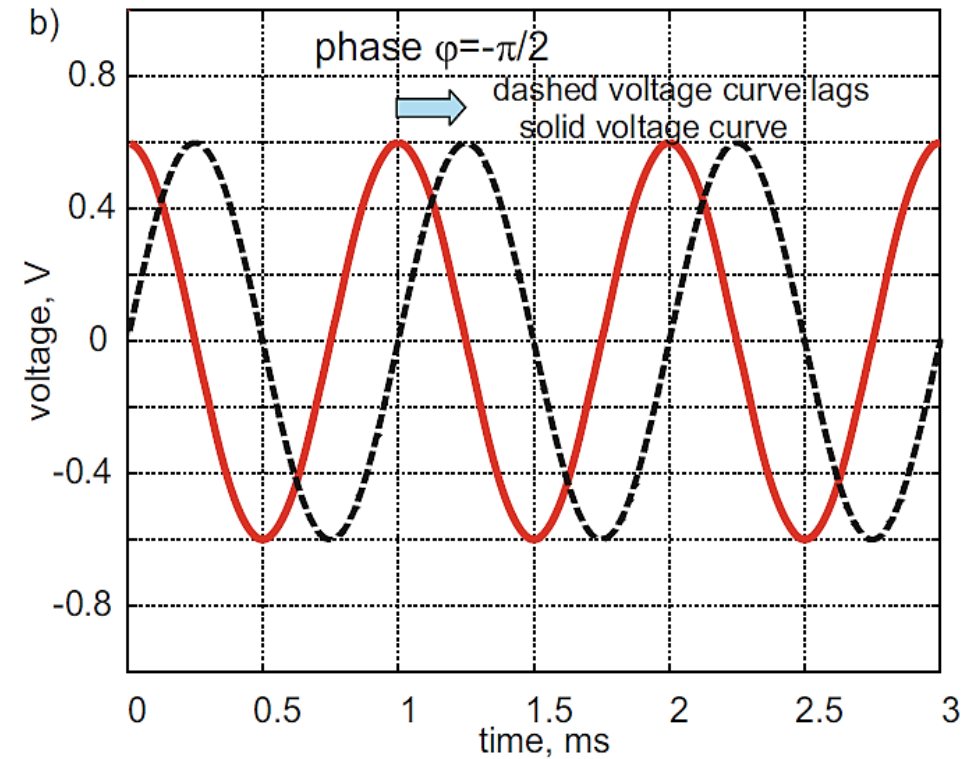
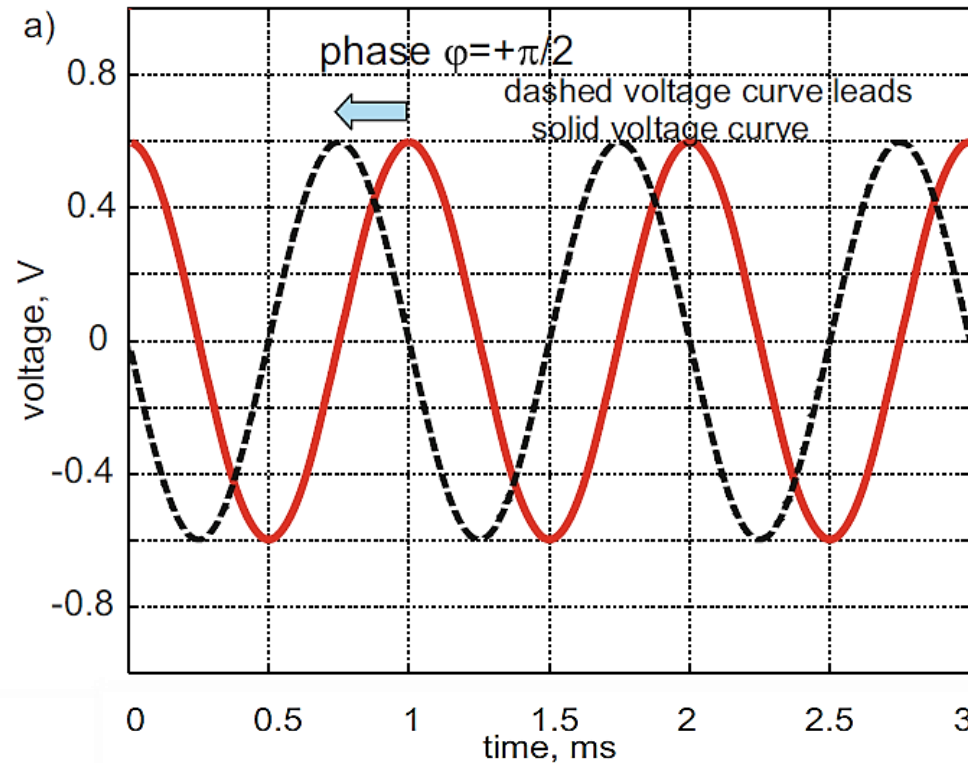


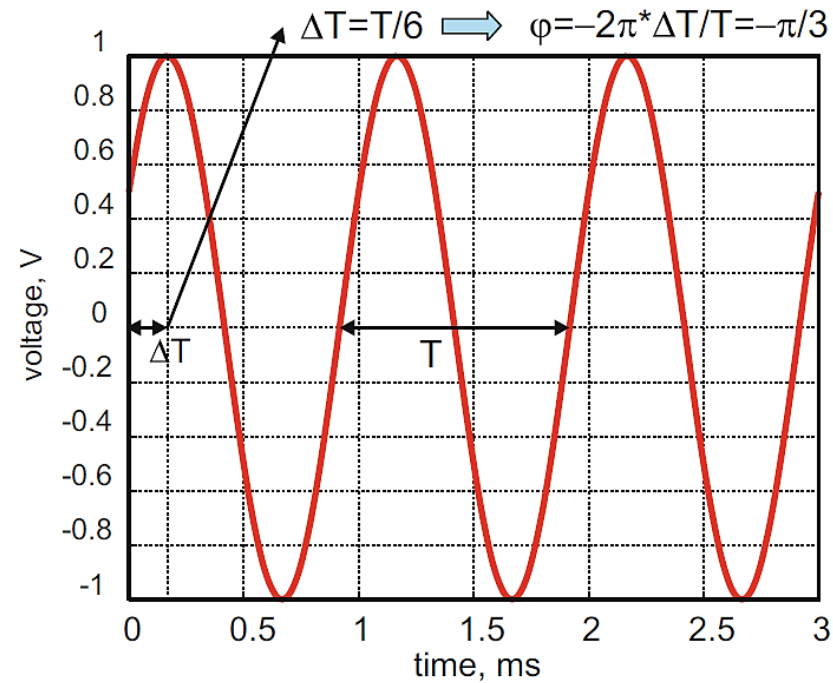
SOLVED NUMERICALS ON SINUSOIDS

Numerical 1: Determine frequency in Hz, angular frequency in rad/s, and the amplitude of the harmonic voltage signals as shown.



Solution: All four signals in Fig. have the same **amplitude** of **0.6 V**, the same **frequency** of $f = 1 \text{ kHz}$, and the same **angular frequency** of $\omega = 2\pi f = 6283.1 \text{ rad/s}$.

Numerical 2: Determine the frequency, amplitude, and phase of the harmonic voltage signal as shown versus the base cosine signal.

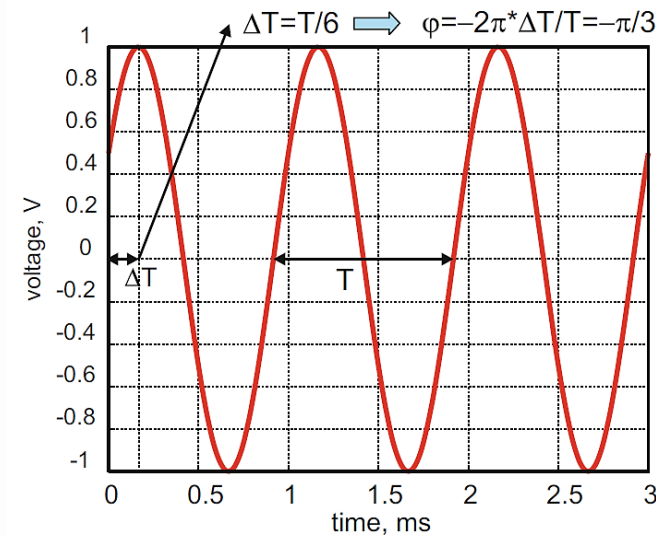


Solution: The amplitude by inspection is observed to be $V_m = 1.0\text{V}$. The frequency is determined by observing the entire interval from 0 to 3 ms containing three full periods; hence $T = 1\text{ ms}$, and $f = 1/T = 1000\text{ Hz} = 1\text{ kHz}$.

For the phase determination, we note that the first maximum occurs later in time than for the base cosine, which already peaks at $t = 0$.

Therefore, the phase must be negative, that is, $\varphi < 0$. The absolute value of the phase is,

$$|\varphi| = 2\pi \frac{\Delta T}{T}$$



which gives $|\varphi| = \pi/3$ for $\Delta T = T/6$

Alternatively, the same result can be obtained by observing that the cosine function is equal to $0.5 V_m$ at $t = 0$, so that $\varphi = -\cos^{-1}(0.5) = -\pi/3$.

Numerical 3: Find the amplitude, phase, period, and frequency of the sinusoid $v(t) = 12 \cos(50t + 10^\circ)$ V.

Solution:

The amplitude is $V_m = 12$ V

The phase is $\phi = 10^\circ$

The angular frequency is $\omega = 50$ rad/s

The period $T = 2\pi/\omega = 2\pi/50 = 0.1257$ s

The frequency is $f = 1/T = 7.958$ Hz

Numerical 4: Given the sinusoid $45 \cos(5\pi t + 36^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Solution: 45, 36° , 15.708 rad/s, 400 ms, 2.5 Hz

Numerical 5: Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

Solution: METHOD 1- In order to compare v_1 and v_2 , we must express them in the same form. If we express them in cosine form with positive amplitudes, $v_1 = -10 \cos(\omega t + 50^\circ) = 10 \cos(\omega t + 50^\circ - 180^\circ)$
 $v_1 = 10 \cos(\omega t - 130^\circ)$ or $v_1 = \mathbf{10 \cos(\omega t + 230^\circ)}$

$$v_2 = 12 \sin(\omega t - 10^\circ) = 12 \cos(\omega t - 10^\circ - 90^\circ)$$
$$v_2 = 12 \cos(\omega t - 100^\circ) = 12 \cos(\omega t - 100^\circ + 360^\circ) = \mathbf{12 \cos(\omega t + 260^\circ)}$$

It can be deduced that the phase difference between v_1 and v_2 is 30°
Thus, it can be observed v_2 **leads** v_1 **by** 30°

METHOD 2- Alternatively, we may express v_1 in sine form:

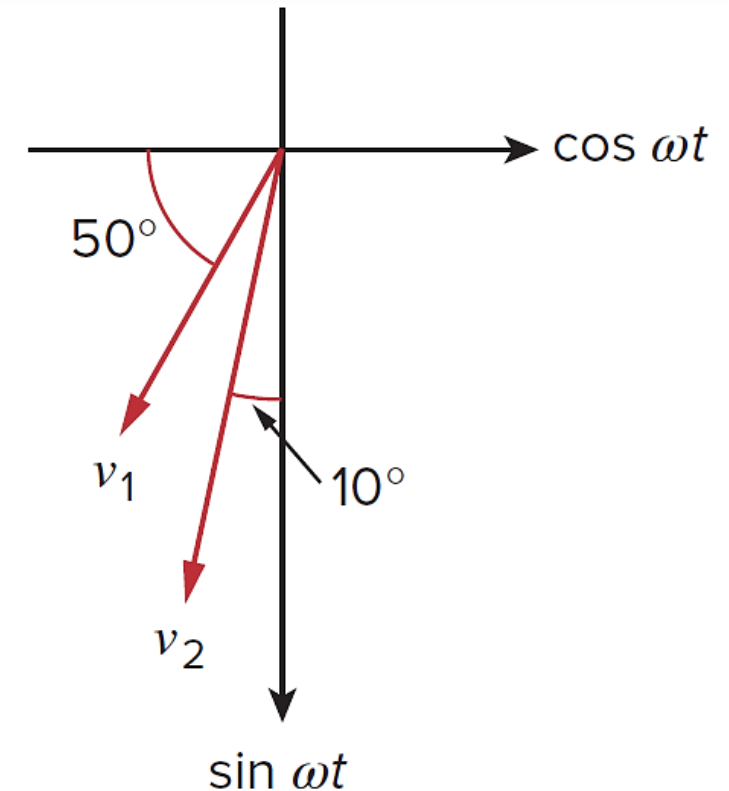
$$\begin{aligned} v_1 &= -10 \cos(\omega t + 50^\circ) = 10 \sin(\omega t + 50^\circ - 90^\circ) \\ &= 10 \sin(\omega t - 40^\circ) = \mathbf{10 \sin(\omega t - 10^\circ - 30^\circ)} \end{aligned}$$

But $v_2 = 12 \sin(\omega t - 10^\circ)$. Comparing the two shows that v_1 **lags v_2 by 30°** . This is the same as saying that v_2 **leads v_1 by 30°**

METHOD 3- We may regard v_1 as simply $-10 \cos \omega t$ with a phase shift of $+50^\circ$.

Similarly, v_2 is $12 \sin \omega t$ with a phase shift of -10° .

Therefore, as observed v_2 **leads v_1 by 30°** .



BASICS OF PHASORS

BASICS OF PHASORS

A **phasor** is a complex number that represents the amplitude and phase of a sinusoid.

- ❑ Phasors provide a simple means of analysing linear circuits excited by sinusoidal sources
- ❑ A complex number z can be written in rectangular form as

$$z = x + jy$$

where $j = \sqrt{-1}$, x is the real part of z , y is the imaginary part of z

BASICS OF PHASORS

- ❑ The complex number z can also be written in polar or exponential form as,

$$z = r \angle \phi = re^{j\phi}$$

Where, r is the magnitude of z , and ϕ is the phase of z

$$z = x + jy,$$

Rectangular form

$$z = r \angle \phi$$

Polar form

- ❑ The relationship between the rectangular form and the polar form is shown in Fig.,

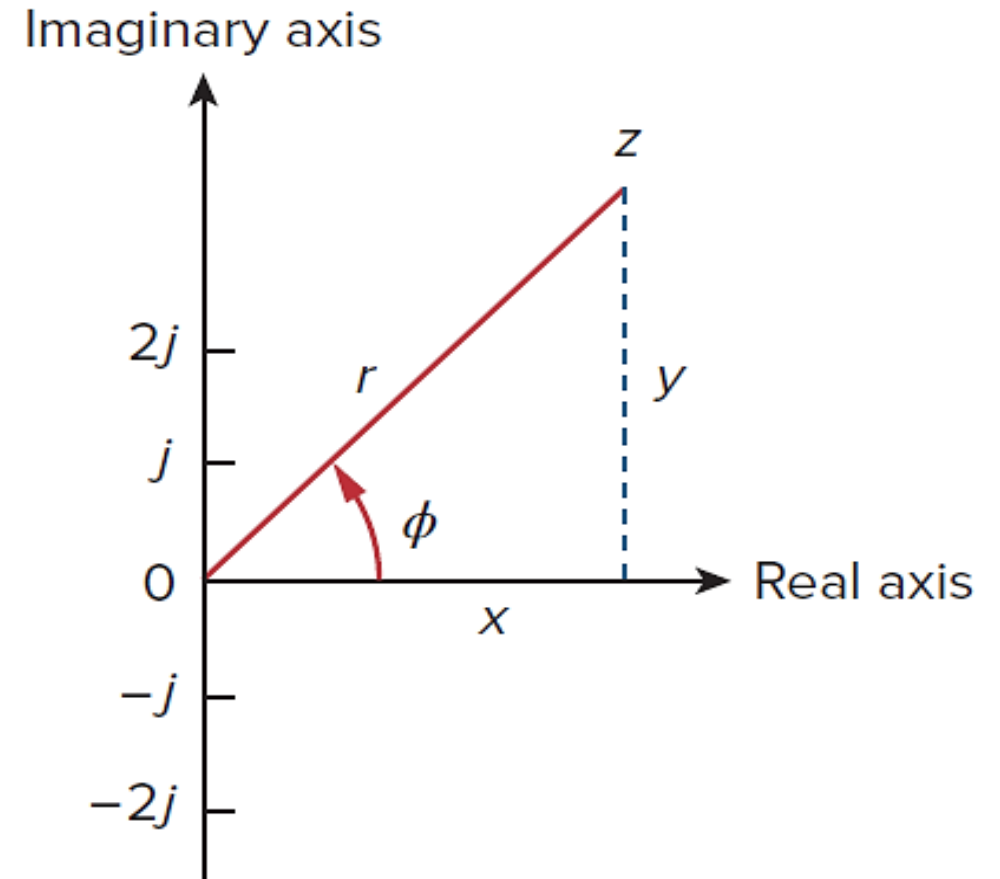


Fig. Representation of a complex number

BASICS OF PHASORS

- ❑ Where, x axis represents the real part and the y axis represents the imaginary part of a complex number
- ❑ Given x and y , we can get r and ϕ as

$$r = \sqrt{x^2 + y^2}, \phi = \tan^{-1} \frac{y}{x} \quad \text{where, } x = r \cos \phi, y = r \sin \phi$$

- ❑ Thus, z may be written as

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi)$$

- ❑ The idea of phasor representation is based on Euler's identity

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$

BASICS OF PHASORS

□ By considering $\cos \phi$ and $\sin \phi$ as the real and imaginary parts of $e^{j\phi}$,

$$\cos \phi = \operatorname{Re}(e^{j\phi}), \sin \phi = \operatorname{Im}(e^{j\phi})$$

□ Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$,

$$v(t) = V_m \cos(\omega t + \phi) = \operatorname{Re}(V_m e^{j(\omega t + \phi)})$$

□ Thus, $v(t) = \operatorname{Re}(V e^{j\omega t})$ where, $V = V_m e^{j\phi} = V_m \angle \phi$

BASICS OF PHASORS

❑ The plot of $\mathbf{V}e^{j\omega t} = V_m e^{j(\omega t + \phi)}$ on the complex plane can be observed as,

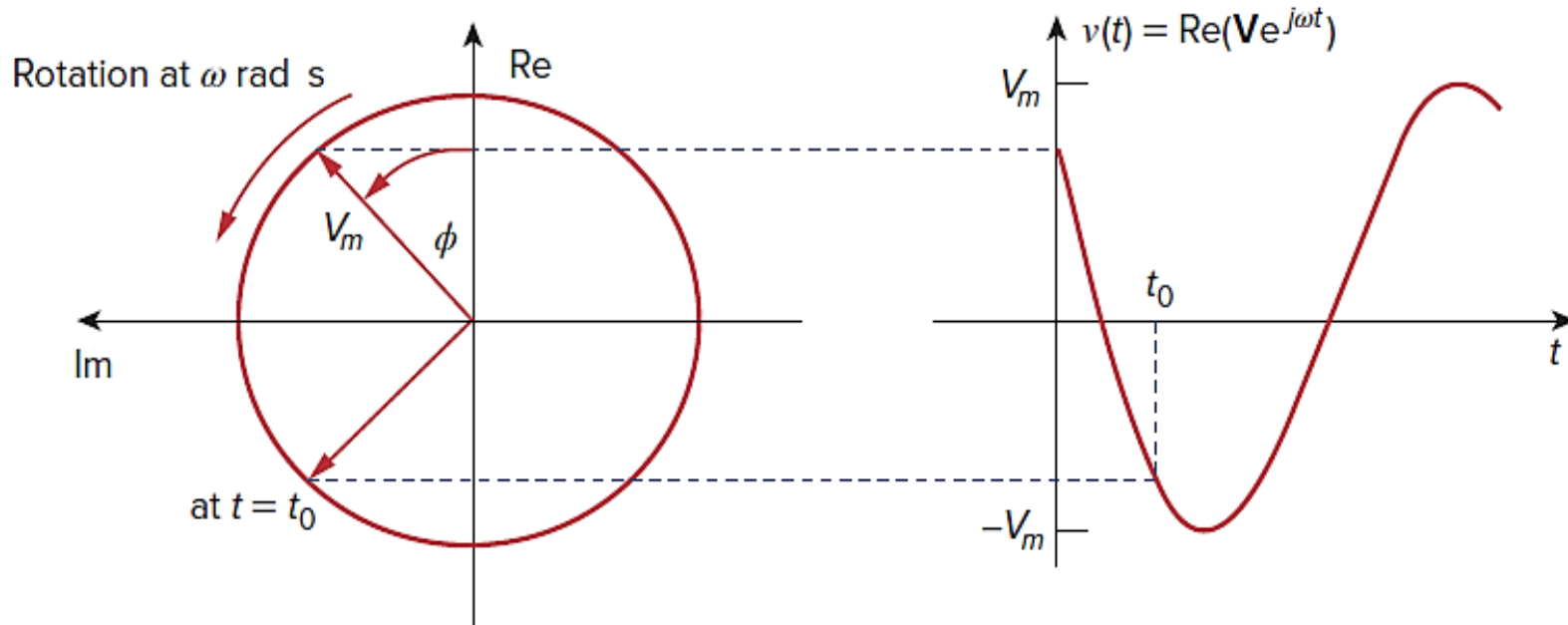


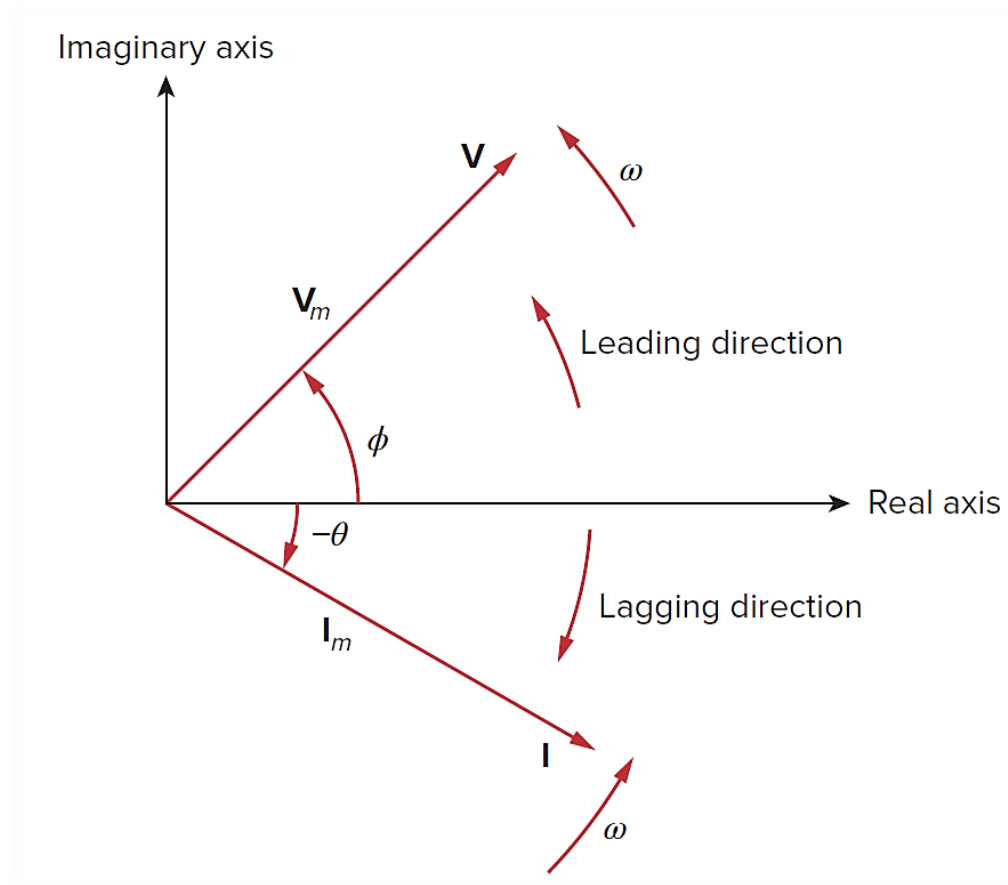
Fig. Representation of $\mathbf{V}e^{j\omega t}$ sinor rotating counter clockwise and its projection on the real axis as a function of time.

❑ As time increases, the sinor rotates on a circle of radius V_m at an angular velocity ω in the counter clockwise direction

BASICS OF PHASORS

❑ Phasors $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$ are graphically represented,

Such a graphical representation of phasors is known as a *phasor diagram*



BASICS OF PHASORS

- ❑ By suppressing the time factor, we transform the sinusoid from the time domain to the phasor domain. This transformation is summarized as follows,

$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow V = V_m \angle \phi$$

- ❑ Besides time differentiation and integration, another important use of phasors is found in summing sinusoids of the same frequency

SOLVED NUMERICALS ON PHASORS

Numerical 1: Transform these sinusoids to phasors,

(a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

(b) $v = -4 \sin(30t + 50^\circ) \text{ V}$

Solution:

(a) $i = 6 \cos(50t - 40^\circ) \text{ A}$ has the phasor, $I = 6 \angle -40^\circ \text{ A}$

(b) Since, $-\sin A = \cos(A + 90^\circ)$,

$$v = -4 \sin(30t + 50^\circ) = 4 \cos(30t + 50^\circ + 90^\circ) = 4 \cos(30t + 140^\circ) \text{ V}$$

The phasor form of v is $V = 4 \angle 140^\circ \text{ V}$

Numerical 2: Express these sinusoids as phasors,

(a) $v = -14 \sin(5t - 22^\circ) \text{ V}$

(b) $i = -8 \cos(16t + 15^\circ) \text{ A}$

Solution: (a) $V = 14 \angle 68^\circ \text{ V}$ (b) $I = 8 \angle -165^\circ \text{ A}$

Numerical 3: Find the sinusoids corresponding to these phasors,

(a) $V = -25 \angle 40^\circ \text{ V}$

(b) $I = j(12 - j5) \text{ A}$

Solution: (a) $v(t) = 25 \cos(\omega t - 140^\circ) \text{ V}$ or $25 \cos(\omega t + 220^\circ) \text{ V}$,

(b) $i(t) = 13 \cos(\omega t + 67.38^\circ) \text{ A}$

Numerical 4: Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ A and $i_2(t) = 5 \sin(\omega t - 20^\circ)$ A, find their sum.

Solution:

Current $i_1(t)$ is in the standard form. Its phasor is $I = 4 \angle 30^\circ$ A

We need to express $i_2(t)$ in cosine form. The rule for converting sine to cosine is to subtract by 90° ,

$$i_2 = 5 \cos(\omega t - 20^\circ - 90^\circ) = 5 \cos(\omega t - 110^\circ)$$

and its phasor is $I_2 = 5 \angle -110^\circ$ A

If we let $i = i_1 + i_2$, then

$$\begin{aligned}\mathbf{I} &= \mathbf{I}_1 + \mathbf{I}_2 = 4\angle 30^\circ + 5\angle -110^\circ \text{ A} \\ &= (3.464 + j2) + (-1.71 - j4.698) \\ &= 1.754 - j2.698 \\ &= 3.218\angle -56.97^\circ \text{ A}\end{aligned}$$

Transforming this to the time domain,
we get $i(t) = 3.218 \cos(\omega t - 56.97^\circ) \text{ A}$

AVERAGE AND RMS VALUES OF AN ALTERNATING CURRENT

AVERAGE AND RMS VALUES OF AN ALTERNATING CURRENT

- Let us first consider the wave shown as shown, which is typical current waveform produced by a transformer on no load,

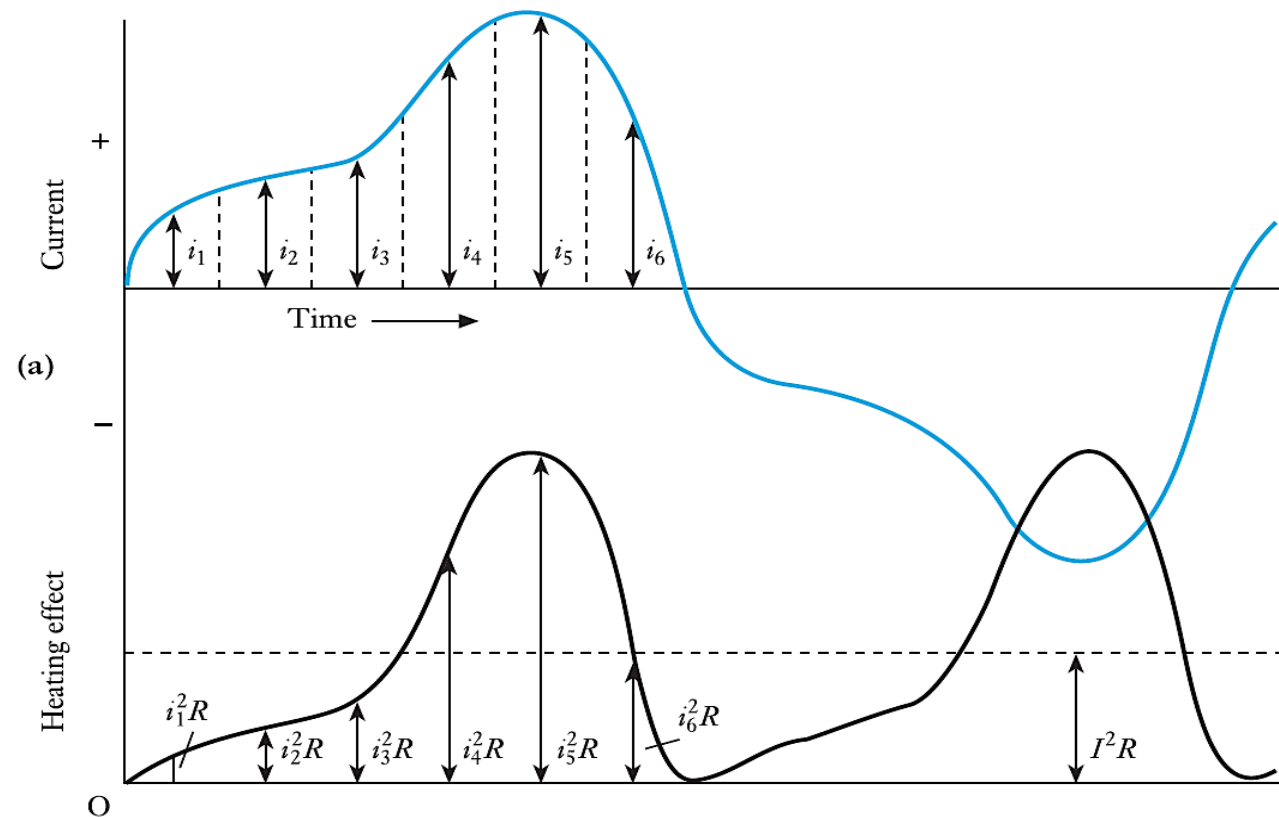


Fig. Average and RMS values

AVERAGE VALUE OF AN ALTERNATING CURRENT

- If n equidistant mid-ordinates ($i_1, i_2, \dots$) are taken over either the positive or the negative half-cycle, then *average* value of current over half a cycle is,

$$I_{av} = \frac{i_1 + i_2 + \dots + i_n}{n}$$

Or, alternatively, average value of current is,

$$= \frac{\text{Area enclosed over half - cycle}}{\text{Length of base over half - cycle}}$$

AVERAGE AND RMS VALUES OF AN ALTERNATING CURRENT

- ❑ If I_m is the maximum value of a current which varies sinusoidally as shown,

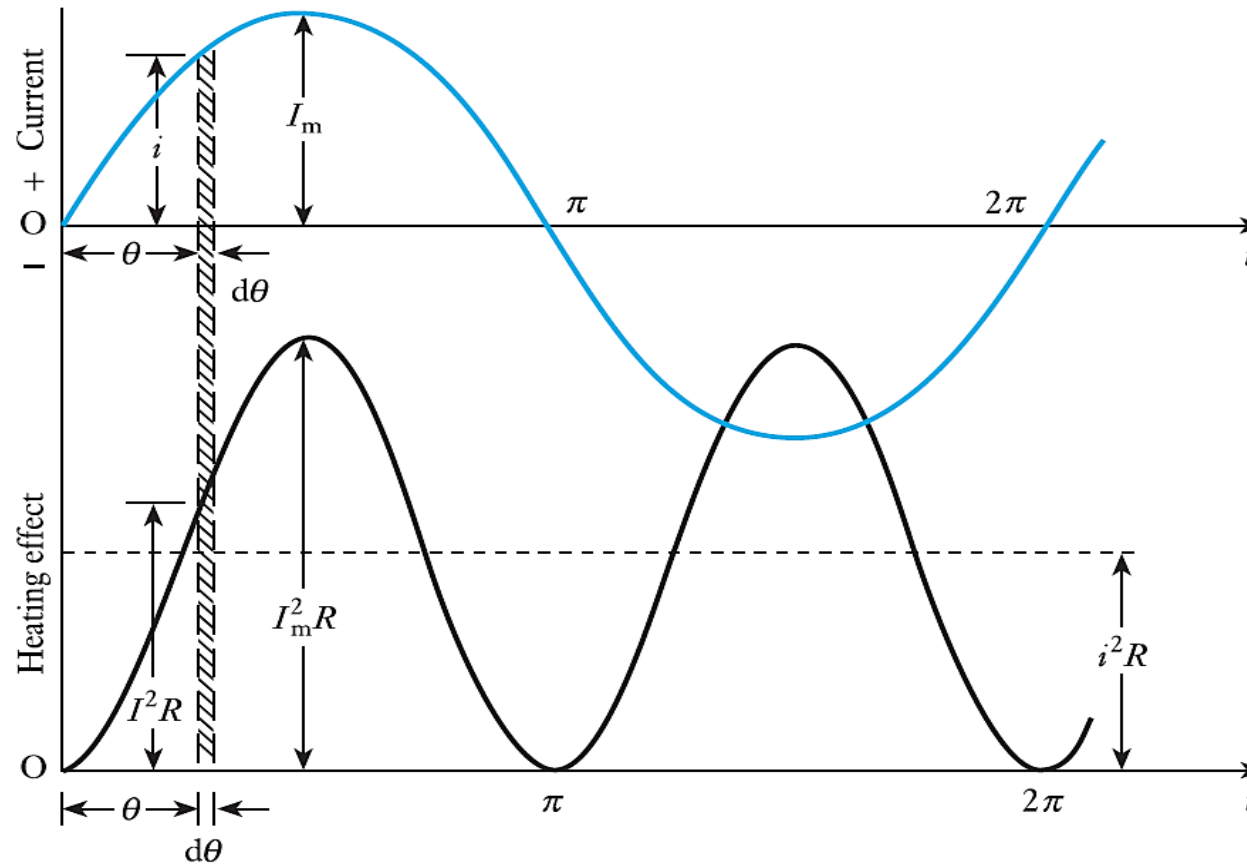


Fig. Average and RMS values of a sinusoidal current

AVERAGE AND RMS VALUES OF AN ALTERNATING CURRENT

- ❑ The instantaneous value i is represented by,

$$i = I_m \sin \theta$$

where, θ is the angle in radians from instant of zero current

- ❑ For a very small interval $d\theta$ radians, the area of the shaded strip is
 $i \cdot d\theta$ ampere radians
- ❑ The use of the unit '*ampere radian*' avoids converting the scale on the horizontal axis from radians to seconds

AVERAGE VALUE OF AN ALTERNATING CURRENT

□ Therefore, total area enclosed by the current wave over half-cycle is,

$$\begin{aligned}\int_0^{\pi} i \cdot d\theta &= I_m \int_0^{\pi} \sin \theta \cdot d\theta = -I_m [\cos \theta]_0^{\pi} \\ &= -I_m [(-1) - 1] = 2I_m\end{aligned}$$

□ Therefore, average value of current over a half-cycle is $I_{av} = \frac{2I_m}{\pi}$

$$I_{av} = 0.637I_m$$

RMS VALUE OF AN ALTERNATING CURRENT

□ If i is the instantaneous current through the resistance, the average power dissipated is,

$$\langle i^2 R \rangle = I_{RMS}^2 R \quad \text{where } \sqrt{\langle i^2 \rangle} = I_{RMS}$$

$$\langle i^2 \rangle = \frac{1}{\pi} \int_0^\pi I_m^2 \sin^2 \theta d\theta = \frac{I_m^2}{2}$$

Therefore, $I_{RMS} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$

It is normal practice to omit the RMS subscript, and just denote I_{RMS} as I

AVERAGE AND RMS VALUES OF AN ALTERNATING CURRENT

- ❑ RMS value of a sinusoidal current or voltage is,

$$I = 0.707 I_m$$

- ❑ Form factor (= RMS/Average) of a sine wave is,

$$\text{Form factor} = \frac{0.707 * \text{maximum value}}{0.637 * \text{minimum value}}$$

$$k_f = 1.11$$

- ❑ Peak or crest factor (= Amplitude/Average) of a sine wave is

$$\text{Peak factor} = \frac{\text{maximum value}}{0.707 * \text{maximum value}}$$

$$k_p = 1.414$$

NUMERICALS ON *AVERAGE* AND RMS VALUES

Numerical 1: An alternating current of sinusoidal waveform has an RMS value of 10.0 A. What is the peak-to-peak value of this current?

Solution: $I_m = \frac{I}{0.707} = \frac{10}{0.707} = 14.14A$

The peak-to-peak value is therefore **14.14 – (-14.14) = 28.28 A**

Numerical 2: An alternating voltage has the equation $v = 141.4 \sin 377t$, what are the values of (a) RMS voltage (b) frequency (c) the instantaneous voltage when $t = 3 \text{ ms}$?

Solution: The relation is of the form $v = V_m \sin \omega t$ and by comparison,

$$(a) V_m = 141.4V = \sqrt{2}V \quad \text{Hence, } V = \frac{141.4}{\sqrt{2}} = 100V$$

(b) Also by comparison,

$$\omega = 377 \text{ rad / s} = 2\pi f, \quad f = \frac{377}{2\pi} = 60 \text{ Hz}$$

(c) Finally, $v = 141.4 \sin 377t$

when $t = 3 \times 10^{-3} \text{ sec}$,

$$\begin{aligned} v &= 141.4 \sin(377 \times 3 \times 10^{-3}) = 141.4 \sin 1.131 \\ &= 141.4 \times 0.904 = 127.8V \end{aligned}$$

Numerical 3: A current has the following steady values in amperes for equal intervals of time changing instantaneously from one value to the next (Fig. 32):

0, 10, 20, 30, 20, 10, 0, -10, -20, -30, -20, -10, 0, etc.

Calculate the RMS value of the current and its form factor.

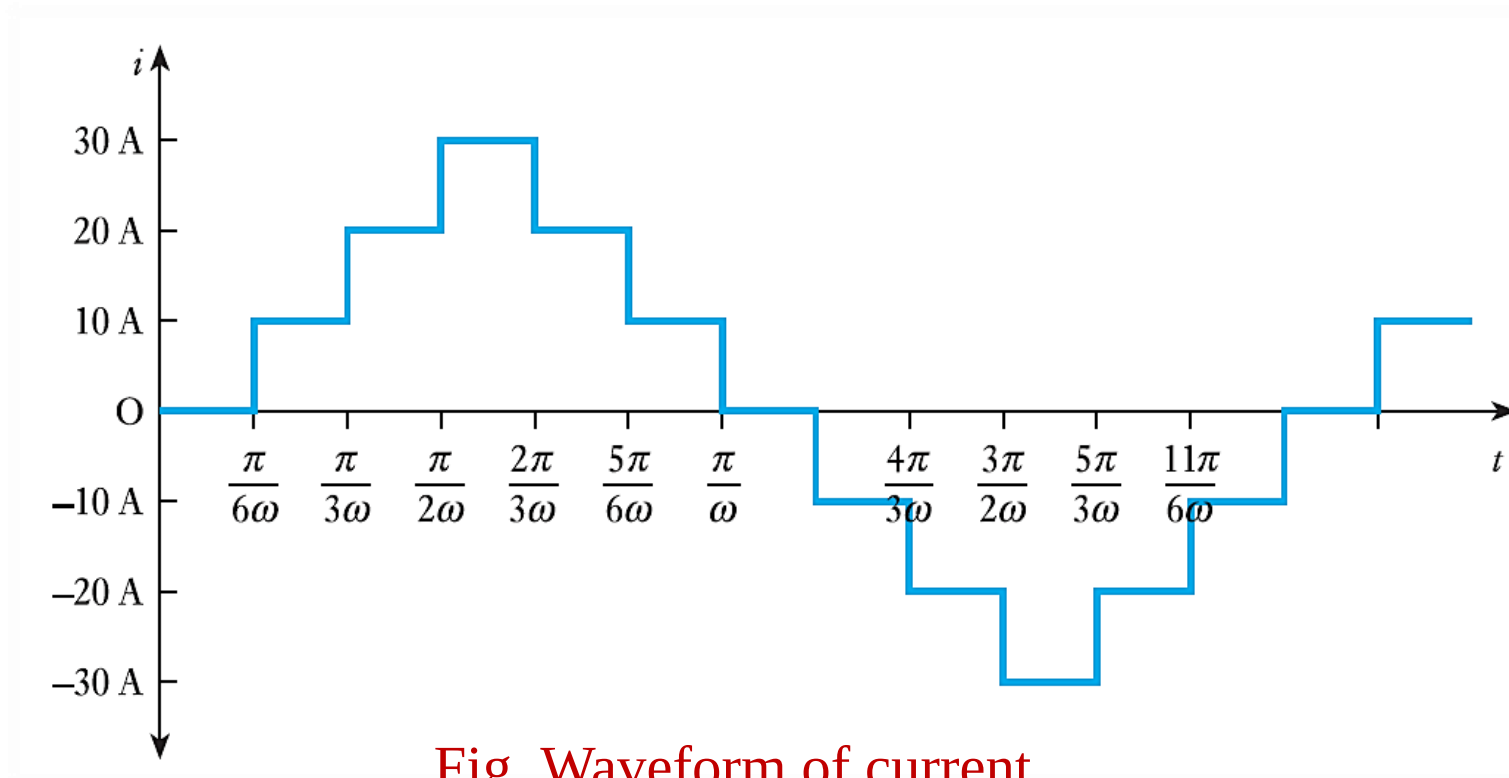


Fig. Waveform of current

Solution:

$$I_{av} = \frac{\text{area under curve}}{\text{length of base}}$$

$$= \frac{0 \left[\frac{\pi}{6\omega} - 0 \right] + 10 \left[\frac{2\pi}{6\omega} - \frac{\pi}{6\omega} \right] + 20 \left[\frac{3\pi}{6\omega} - \frac{2\pi}{6\omega} \right] + 30 \left[\frac{4\pi}{6\omega} - \frac{3\pi}{6\omega} \right] + 20 \left[\frac{5\pi}{6\omega} - \frac{4\pi}{6\omega} \right] + 10 \left[\frac{6\pi}{6\omega} - \frac{5\pi}{6\omega} \right]}{\frac{\pi}{\omega} - 0}$$

$$= 15A$$

$$I^2 = \frac{0^2 \left[\frac{\pi}{6\omega} - 0 \right] + 10^2 \left[\frac{2\pi}{6\omega} - \frac{\pi}{6\omega} \right] + 20^2 \left[\frac{3\pi}{6\omega} - \frac{2\pi}{6\omega} \right] + 30^2 \left[\frac{4\pi}{6\omega} - \frac{3\pi}{6\omega} \right] + 20^2 \left[\frac{5\pi}{6\omega} - \frac{4\pi}{6\omega} \right] + 10^2 \left[\frac{6\pi}{6\omega} - \frac{5\pi}{6\omega} \right]}{\frac{\pi}{\omega} - 0}$$

$$= 316$$

$$I = \sqrt{316} = 17.8A$$

$$k_f = \frac{I}{I_{av}} = \frac{17.8}{15.0} = 1.19$$